AI QB UNIT 5 (4 marks)

Q1. Define the planning problem in AI.

1. A planning problem in Artificial Intelligence is the process of finding a sequence of actions to achieve a specific goal. It helps an intelligent agent decide what steps should be performed to move from the initial state to the goal state.
2. In AI planning, the system first studies the current situation and then identifies the desired target or objective. After this, it generates a suitable plan that can successfully achieve the required result.
3. A planning problem mainly consists of three important parts which are the initial state, goal state, and available actions. The initial state describes the starting condition, while the goal state defines the final condition to be achieved.
4. Actions are operations that change one state into another state during the planning process. Each action has certain preconditions and effects that determine whether the action can be applied or not.
5. Planning problems are widely used in robotics, navigation systems, game playing, and automated scheduling applications. AI systems use planning techniques to make intelligent decisions in complex environments.
6. The main objective of planning is to find the best and most efficient path toward the goal. A good planning system reduces time, cost, and unnecessary actions while solving a problem.

Q2. What is state space search in planning?

1. State space search in planning is a method where all possible states of a problem are explored to find a solution. It helps an AI system move step by step from the initial state to the goal state.
2. In this method, each state represents a particular condition or situation of the problem. The system applies different actions to generate new states until the desired goal is achieved.
3. State space search can be performed using forward search or backward search techniques. Forward search starts from the initial state, while backward search starts from the goal state.
4. The search process forms a graph or tree structure where nodes represent states and edges represent actions. The AI system explores these nodes systematically to identify the correct path.
5. This technique is commonly used in puzzle solving, robot movement, navigation systems, and game playing applications. Algorithms like Breadth First Search and Depth First Search are often used for state space exploration.
6. The main advantage of state space search is that it provides a clear and organized way to solve planning problems. However, large problems may produce a huge number of states which increases memory and computation requirements.

Q3. Define partial order planning.

1. Partial Order Planning is a planning technique in Artificial Intelligence where actions are arranged only when necessary. Instead of fixing the complete sequence, the planner allows some actions to remain unordered.
2. This method follows the principle of least commitment which means decisions are delayed until required. It gives more flexibility during the planning process and avoids unnecessary restrictions.
3. In partial order planning, actions are connected using ordering constraints and causal links. These constraints ensure that important actions occur in the correct order to achieve the goal successfully.
4. The planner starts with an initial plan and gradually adds actions, conditions, and constraints. It continues refining the plan until all goals and subgoals are satisfied properly.
5. Partial order planning is useful in complex problems where multiple tasks can be performed independently or simultaneously. It is commonly applied in robotics, scheduling systems, and automated task management.
6. The main advantage of partial order planning is flexibility because independent actions can occur in different sequences. It also reduces unnecessary commitments and improves planning efficiency in complicated environments.

Q4. What is hierarchical planning?

1. Hierarchical planning is a planning technique in Artificial Intelligence where a large problem is divided into smaller subproblems. The system solves high-level tasks first and then breaks them into detailed lower-level actions.
2. This approach organizes planning into different layers or levels of abstraction. Higher levels focus on general goals while lower levels handle specific operations and execution steps.
3. In hierarchical planning, complex tasks are represented using task decomposition methods. Each major task is divided into smaller manageable subtasks until simple executable actions are obtained.
4. This method helps reduce the complexity of planning because the system does not need to handle all details at the same time. It allows the planner to focus on one level before moving to the next level.
5. Hierarchical planning is widely used in robotics, military planning, manufacturing systems, and game AI applications. It is especially useful for solving large and complicated real-world problems efficiently.
6. The main advantage of hierarchical planning is that it improves planning speed and reduces computational effort. It also makes the planning process more organized, flexible, and easier to understand.

Q5. Define conditional planning.

1. Conditional planning is a type of AI planning where different actions are chosen based on different conditions or situations. The plan changes according to the information received from the environment during execution.
2. In conditional planning, the system prepares alternative paths for possible future events or outcomes. This helps the agent continue functioning properly even when uncertainty exists in the environment.
3. Conditional planning uses condition checks and decision branches to select suitable actions. If one condition becomes true, the planner follows one path, and if another condition occurs, it follows a different path.
4. This planning method is useful when the environment is unpredictable or partially observable. The agent may not know all information in advance, so it must adapt its actions dynamically.
5. Conditional planning is commonly used in robotics, self-driving cars, medical diagnosis systems, and automated control systems. These applications often require decisions based on changing real-world conditions.
6. The main advantage of conditional planning is that it provides flexibility and adaptability in uncertain environments. However, designing multiple conditional paths can increase the complexity of the planning process.

Q6. List the forms of learning in AI.

1. Artificial Intelligence uses different forms of learning to improve system performance through experience and data. These learning methods help intelligent agents solve problems and make better decisions over time.
2. Rote learning is a simple form of learning where the system memorizes facts or previously solved problems. It retrieves stored information directly whenever the same problem appears again.
3. Learning by taking advice allows the system to learn from instructions provided by humans or experts. This method helps the system learn quickly without discovering everything by trial and error.
4. Learning from examples, also called inductive learning, helps the system generate general rules from specific examples. The system studies patterns in the examples and then applies the learned rules to new situations.
5. Learning by analogy solves new problems using similarities with previously solved problems. Explanation-based learning helps the system understand why a solution works instead of only memorizing it.
6. Reinforcement learning is another important form where the agent learns through rewards and punishments from the environment. Discovery learning allows the system to independently discover new rules and concepts through exploration.

Q7. What is reinforcement learning?

1. Reinforcement Learning is a branch of Artificial Intelligence in which an agent learns by interacting with the environment. The agent performs actions and receives rewards or penalties based on its behavior.
2. The main goal of reinforcement learning is to maximize the total reward over time. The agent gradually learns the best strategy or policy through continuous experience and feedback.
3. In reinforcement learning, the agent first observes the current state and then selects an action. After the action is performed, the environment changes and provides a reward or penalty.
4. Positive rewards encourage the agent to repeat good actions, while negative rewards discourage wrong actions. Through repeated trials, the system learns which actions produce better outcomes.
5. Reinforcement learning is based on trial-and-error learning and does not require labeled training data. The agent improves automatically by continuously interacting with the environment.
6. Reinforcement learning is widely used in robotics, game playing, self-driving cars, and recommendation systems. It is especially useful for solving sequential decision-making problems in complex environments.

Q8. Define learning from rewards.

1. Learning from rewards is a method in Artificial Intelligence where an agent learns by receiving feedback from the environment. The feedback is given in the form of rewards for correct actions and penalties for wrong actions.
2. In this learning process, the agent performs different actions and observes their outcomes. Actions that produce positive rewards are encouraged, while actions with negative results are avoided.
3. The main objective of learning from rewards is to maximize long-term benefits or cumulative rewards. The system gradually identifies the best actions through repeated experience and interaction.
4. This type of learning is mainly used in reinforcement learning systems where no direct teacher is available. The agent learns independently by analyzing rewards received after every action.
5. Learning from rewards is useful in robot navigation, gaming systems, self-driving cars, and automated decision-making applications. These systems continuously improve their performance based on reward feedback.
6. The advantage of learning from rewards is that the system can adapt automatically to changing environments. However, designing suitable reward functions can sometimes be difficult and computationally expensive.

Q9. What is statistical learning (basic concept)?

1. Statistical learning is a branch of Artificial Intelligence and Machine Learning that uses statistics and probability to learn from data. It helps systems analyze patterns and make predictions automatically.

2. Statistical learning combines statistical techniques, probability theory, and machine learning methods to process information effectively. The system studies data patterns and uses them for prediction and decision-making.

3. The main goal of statistical learning is to discover useful relationships and hidden patterns from large amounts of data. It is widely used for prediction, pattern recognition, and data analysis tasks.

4. The statistical learning process starts with data collection from sources such as databases, sensors, surveys, or websites. The collected data is then organized into training data for the learning algorithm.

5. Different learning algorithms such as Decision Trees, Neural Networks, and Linear Regression are used to build models from training data. The generated model can then make predictions on new unseen data.

6. Statistical learning is applied in medical diagnosis, fraud detection, recommendation systems, speech recognition, and weather forecasting. It improves system performance by learning continuously from data and experience.

(5 – marks)

Q1. Explain planning with state space search.

1. Planning with state space search is a method in Artificial Intelligence where the system searches through different states to reach a goal. The planner moves step by step from the initial state to the final goal state using suitable actions.
2. In this method, a state represents a particular situation or condition of the problem environment. Actions are applied to the current state to generate new possible states during the search process.
3. State space search mainly uses two approaches called forward search and backward search. Forward search starts from the initial state, while backward search starts from the goal state and moves backward.
4. The search process forms a graph or tree structure where nodes represent states and edges represent actions. The planner explores these nodes systematically until the required goal state is found successfully.
5. Different search algorithms like Breadth First Search, Depth First Search, and A* Search are used in state space planning. These algorithms help the system find efficient and optimal paths to solve problems.
6. State space search is widely used in robotics, navigation systems, puzzle solving, and game playing applications. It helps AI systems make intelligent decisions by exploring possible future situations.
7. The main advantage of planning with state space search is that it provides a structured method for solving complex problems. However, very large state spaces may increase memory usage and computational complexity significantly.

Q2. Explain partial order planning with example.

1. Partial Order Planning is a planning technique in Artificial Intelligence where actions are ordered only when necessary. The planner does not fix the complete sequence of actions at the beginning of the planning process.
2. This method follows the principle of least commitment which means decisions are delayed until required. It allows the planner to keep flexibility while generating a solution plan.
3. In partial order planning, actions are connected using ordering constraints and causal links. These links ensure that actions happen in the correct order whenever dependency exists between them.

4. The planner begins with an incomplete plan and gradually adds actions, conditions, and constraints to satisfy all goals. Independent actions can remain unordered and may be executed in different sequences.
5. For example, while preparing for a trip, packing clothes and charging a mobile phone can be done in any order. However, booking tickets must be completed before starting the journey.
6. Partial order planning is useful in robotics, scheduling systems, automated manufacturing, and task management applications. It is especially effective when many tasks can occur simultaneously or independently.
7. The main advantage of partial order planning is flexibility because it avoids unnecessary restrictions on action order. It also improves efficiency by reducing the complexity of planning in large environments.

Q3. Explain hierarchical planning.

1. Hierarchical planning is a planning method in Artificial Intelligence where a large task is divided into smaller subtasks. The planner first handles high-level goals and then breaks them into detailed lower-level actions.
2. This planning technique organizes tasks into multiple levels of abstraction. Higher levels focus on general objectives, while lower levels focus on specific executable actions.
3. In hierarchical planning, complex problems are simplified through task decomposition. Each major task is divided repeatedly until simple actions that can be directly executed are obtained.
4. The method reduces planning complexity because the system does not process all details at the same time. It allows the planner to solve one level before moving to another level.
5. For example, in robot delivery planning, the high-level task may be “deliver package to customer.” This task can then be divided into smaller tasks such as picking the package, moving to location, and handing over the package.
6. Hierarchical planning is commonly used in robotics, military operations, manufacturing systems, and game AI applications. It is very useful for solving large and complicated real-world problems efficiently.
7. The major advantage of hierarchical planning is that it improves planning speed and reduces computational effort. It also makes the planning process more organized, manageable, and easier to understand.

Q4. Describe conditional planning with example.

1. Conditional planning is a type of Artificial Intelligence planning where actions depend on different conditions or situations. The planner prepares alternative actions based on possible future outcomes.
2. In conditional planning, the system checks conditions during execution and chooses the appropriate path accordingly. This makes the planning process flexible and adaptable in uncertain environments.
3. The planner creates decision branches for different possible situations that may occur in the future. If one condition becomes true, one set of actions is selected, while another condition leads to a different action sequence.
4. Conditional planning is useful when the environment is unpredictable or partially observable. The system may not know all information in advance, so it must adapt dynamically during execution.
5. For example, a self-driving car may slow down if rain is detected and take another route if traffic is heavy. The car changes its actions based on the current environmental conditions.
6. This planning method is widely used in robotics, healthcare systems, automated control systems, and intelligent navigation systems. These applications often require continuous decision-making under changing conditions.
7. The main advantage of conditional planning is that it improves flexibility and reliability in uncertain environments. However, creating multiple conditional branches can increase the complexity of the planning system.

Q5. Explain theory of learning in AI.

1. Theory of learning in Artificial Intelligence studies how intelligent systems acquire knowledge and improve performance through experience. It explains the principles and methods that make machine learning possible in AI systems.
2. The learning theory mainly focuses on how machines learn correctly from limited data or experience. It also studies how learning algorithms make predictions and decisions accurately.
3. Important elements of learning theory include training data, hypothesis, target function, error measure, and generalization. Training data provides examples for learning, while the hypothesis represents the predicted rule or function.
4. Generalization is an important concept in learning theory where the system performs well on unseen data instead of only memorizing training examples. A

good learner identifies patterns and applies knowledge to new situations successfully.

5. Learning theory also explains bias in learning which refers to assumptions made by learning algorithms. Bias helps reduce complexity, speed up learning, and improve prediction performance.
6. Overfitting and underfitting are major concepts studied in learning theory. Overfitting happens when the model memorizes training data, while underfitting occurs when the model is too simple to learn patterns properly.
7. The main goal of learning theory is to design efficient learning systems that require less data, time, and memory. It helps improve the reliability and performance of AI systems in real-world applications.

Q6. Explain PAC learning concept.

1. PAC learning stands for Probably Approximately Correct learning and is an important concept in machine learning theory. It was introduced by Leslie Valiant to study whether machines can learn efficiently from limited examples.
2. The main idea of PAC learning is that the learner should produce a nearly correct hypothesis with high probability. The system does not need perfect accuracy but should achieve sufficiently good results.
3. In PAC learning, “Probably” means the learning algorithm succeeds with high probability. “Approximately Correct” means the learned solution may contain a small acceptable error.
4. PAC learning uses important terms such as concept class, hypothesis, error, epsilon, and delta. Epsilon represents the maximum acceptable error, while delta represents the maximum probability of failure.
5. A hypothesis is considered PAC-learned if its error is less than or equal to epsilon and the failure probability is less than or equal to delta. This allows small mistakes while still achieving reliable learning performance.
6. PAC learning provides a mathematical framework for analyzing the efficiency and learnability of machine learning algorithms. It helps compare different learning methods and understand their theoretical performance.
7. The major advantage of PAC learning is that it supports statistical learning and efficient model evaluation. However, real-world problems may not always satisfy the strict mathematical assumptions required in PAC theory.

Q7. Explain reinforcement learning framework.

1. Reinforcement Learning is a framework in Artificial Intelligence where an agent learns by interacting with the environment. The agent performs actions and receives rewards or penalties based on its behavior.
2. The reinforcement learning framework mainly consists of an agent, environment, states, actions, and rewards. The agent observes the environment and selects actions to achieve maximum long-term rewards.
3. The process begins when the agent observes the current state of the environment. Based on the observed state, the agent chooses an action and performs it.
4. After the action is executed, the environment changes into a new state and provides feedback in the form of a reward or penalty. Positive rewards encourage good actions while negative rewards discourage wrong actions.
5. The agent continuously repeats this interaction process and gradually improves its strategy or policy over time. Through trial and error learning, the system identifies the best actions for different situations.
6. Reinforcement learning does not require labeled training data or human supervision during learning. It is widely used in robotics, self-driving cars, game playing, and recommendation systems.
7. The major advantage of the reinforcement learning framework is that it allows systems to adapt automatically to changing environments. However, reinforcement learning may require large training time and high computational resources.

Q8. Explain passive reinforcement learning.

1. Passive Reinforcement Learning is a type of reinforcement learning where the agent follows a fixed policy. The agent does not choose actions on its own but only learns how good different states are.
2. In passive reinforcement learning, the actions are already predefined by the given policy. The main objective of the agent is to evaluate the policy and estimate the utility of states.
3. The agent interacts with the environment repeatedly and observes rewards received from different states. States that produce higher rewards are assigned higher utility values over time.
4. Passive reinforcement learning does not involve exploration of new actions or strategies. The agent simply learns from the outcomes produced by the fixed sequence of actions.

5. Important methods used in passive reinforcement learning include Direct Utility Estimation, Adaptive Dynamic Programming, and Temporal Difference Learning. These methods help estimate state utilities using rewards and experiences.
6. For example, in a student learning system, the student follows a fixed study schedule and the system evaluates performance after each test. The system learns which states or study conditions produce better marks.
7. The main advantage of passive reinforcement learning is that it is simple and easier to implement compared to active reinforcement learning. However, it cannot discover better policies because the actions are fixed in advance.

Q9. Explain active reinforcement learning.

1. Active Reinforcement Learning is a type of reinforcement learning where the agent learns which actions should be performed. The agent must decide the best policy by interacting with the environment.
2. In active reinforcement learning, there is no fixed policy given to the agent. The agent explores different actions and learns which actions produce maximum long-term rewards.
3. The agent observes the current state of the environment and then selects an action based on experience. After performing the action, it receives rewards or penalties from the environment.
4. Active reinforcement learning involves exploration and exploitation during the learning process. Exploration helps discover new actions, while exploitation uses known actions that already provide good rewards.
5. Common methods used in active reinforcement learning include Q-Learning and SARSA algorithms. These algorithms help the agent learn the quality or value of actions in different states.
6. For example, a robot moving through a maze tries different paths and learns which path reaches the destination quickly. After repeated trials, the robot identifies the shortest and safest route.
7. The main advantage of active reinforcement learning is that it helps the agent learn optimal actions independently. However, it is more complex and may require more time and computational resources than passive reinforcement learning.

(10 - marks)

Q1. Explain the planning problem in AI in detail with components and examples.

1. A planning problem in Artificial Intelligence is the process of finding a sequence of actions that helps achieve a specific goal. The AI system studies the current situation and decides the best steps required to reach the desired result.
2. Planning is mainly used in intelligent agents where decisions must be taken automatically. It helps systems solve problems logically by selecting suitable actions in the correct order.
3. The first component of a planning problem is the initial state which represents the starting condition of the environment. It describes the current situation before any action is performed.
4. The second component is the goal state which defines the final condition that the system wants to achieve. The planning process continues until this target state is reached successfully.
5. Another important component is actions which are operations that change one state into another. Each action has preconditions and effects that determine how the environment changes.
6. Planning problems also include a state space which contains all possible states generated during the planning process. The AI system searches through this state space to identify the best solution path.
7. For example, in robot navigation, the robot plans actions such as moving left, right, forward, or backward to reach a destination. The robot selects actions step by step until the goal location is reached.
8. In game playing, AI systems use planning to determine the best sequence of moves against opponents. Planning helps the system improve performance and make intelligent decisions.
9. Planning is widely used in robotics, scheduling systems, self-driving cars, and automated manufacturing systems. It helps reduce unnecessary actions, time, and computational effort.
10. The main objective of AI planning is to generate efficient and goal-oriented solutions automatically. A good planning system improves decision-making and solves complex real-world problems effectively.

Q2. Explain planning using state space search with diagram and example.

1. Planning using state space search is a method in Artificial Intelligence where the system explores different states to find a path from the initial state to the goal state. The planner applies actions repeatedly to generate new states during the search process.

2. In this method, a state represents a particular condition or situation of the problem. Actions transform the current state into another state until the required goal is achieved.

3. State space search mainly uses two approaches called forward search and backward search. Forward search starts from the initial state, while backward search begins from the goal state.

4. The planning process forms a graph or tree structure where nodes represent states and edges represent actions. The AI system explores these nodes systematically to identify the correct solution path.

5. A simple diagram of state space search is shown below. The planner moves from the start state toward the goal state through different intermediate states.

Initial State → State A → State B → Goal State
    \→ State C → State D

6. For example, in a maze-solving problem, a robot starts from an initial position and explores different paths. The robot continues selecting actions until it reaches the target location successfully.

7. Different search algorithms such as Breadth First Search, Depth First Search, and A* Search are used in state space planning. These algorithms help identify efficient and optimal solution paths.

8. State space search is widely used in robotics, game playing, navigation systems, and puzzle-solving applications. It provides a systematic way to solve planning problems.

9. The major advantage of state space search is that it offers a clear and organized problem-solving approach. However, large state spaces may increase memory usage and computational complexity.

10. Planning with state space search helps intelligent systems make better decisions by analyzing future possibilities. It improves the ability of AI systems to solve complex tasks effectively.

Q3. Describe partial order planning in detail with example.

1. Partial Order Planning is a planning technique in Artificial Intelligence where actions are ordered only when necessary. The planner avoids fixing the complete sequence of actions at the beginning.
2. This method follows the principle of least commitment which means decisions are delayed until they are actually required. It provides flexibility during the planning process.
3. In partial order planning, actions are connected using ordering constraints and causal links. These constraints ensure that dependent actions occur in the correct order.
4. The planner starts with an incomplete plan that contains only the initial and goal states. Additional actions and constraints are added gradually until all goals are satisfied.
5. Independent actions can remain unordered because their sequence does not affect the final result. This flexibility allows multiple tasks to be performed simultaneously or in different orders.
6. For example, while preparing for a trip, packing clothes and charging a mobile phone can be done independently. However, booking travel tickets must happen before starting the journey.
7. Partial order planning is useful in robotics, manufacturing systems, scheduling applications, and automated task management. It helps manage complex environments with many independent tasks.
8. This planning method reduces unnecessary restrictions on actions and improves efficiency. It also simplifies planning in situations where tasks do not depend on one another.
9. The main advantage of partial order planning is flexibility because the planner commits only to essential action sequences. It also reduces computational complexity in large planning problems.
10. However, partial order planning may become difficult when many dependencies and constraints exist between actions. Despite this, it remains an important planning method in modern AI systems.

Q4. Discuss forms of learning in AI with examples.

1. Artificial Intelligence uses different forms of learning to improve system performance through experience and data. These learning methods help intelligent systems make decisions, solve problems, and adapt to changing environments.
2. Rote learning is a simple form of learning where the system memorizes facts, rules, or previously solved problems. For example, a dictionary lookup system stores meanings of words and retrieves them directly when required.
3. Learning by taking advice allows the system to learn from instructions provided by humans or experts. For example, an expert may provide rules to a medical diagnosis system for identifying diseases.
4. Learning from examples, also called inductive learning, helps the system generate general rules from specific examples. For example, a spam filter studies examples of spam emails and learns patterns for future detection.
5. Learning by analogy solves new problems using similarities with previously solved problems. For example, a robot that knows how to navigate one building may use similar strategies in another building.
6. Explanation-Based Learning helps the system understand why a solution works instead of only memorizing it. For example, a student understands why a mathematical formula works rather than simply remembering the answer.
7. Discovery learning is a form where the system independently discovers new rules or concepts through exploration. Scientific AI systems discovering relationships in experimental data are examples of discovery learning.
8. Reinforcement learning allows an agent to learn through rewards and punishments received from the environment. For example, self-driving cars learn safe driving behavior through continuous interaction with road conditions.
9. Different forms of learning are useful for different types of AI problems and applications. Some methods focus on memorization while others focus on reasoning, exploration, or decision-making.
10. The main goal of learning in AI is to improve system performance automatically over time. These learning methods make AI systems more intelligent, adaptive, and capable of solving complex real-world problems.

Q5. Explain PAC learning theory in detail.

1. PAC learning stands for Probably Approximately Correct learning and is an important concept in machine learning theory. It was introduced by Leslie Valiant to study whether machines can learn efficiently from limited training examples.
2. The main idea of PAC learning is that a learning algorithm should produce a nearly correct hypothesis with high probability. The learner does not need perfect accuracy but should provide sufficiently good results.
3. In PAC learning, “Probably” means the learning system succeeds with high probability. “Approximately Correct” means the hypothesis may contain a small acceptable amount of error.
4. PAC learning provides a mathematical framework for analyzing learnability in machine learning systems. It helps determine under what conditions a concept can be learned efficiently.
5. Important terms used in PAC learning include concept class, hypothesis, target function, error, epsilon, and delta. The concept class represents all possible target concepts that may be learned by the system.
6. Epsilon represents the maximum acceptable error allowed in the learned hypothesis. Delta represents the maximum acceptable probability that the learner may fail.
7. A hypothesis is considered PAC-learned if its error is less than or equal to epsilon and the probability of failure is less than or equal to delta. This means the system achieves reliable learning performance with limited mistakes.
8. PAC learning allows small errors because achieving perfect learning may not always be practical in real-world situations. It focuses on efficient learning instead of absolute correctness.
9. The advantages of PAC learning include providing a strong theoretical foundation for machine learning and helping compare learning algorithms. It also supports statistical learning and performance evaluation.
10. However, PAC learning has limitations because real-world data may not satisfy all mathematical assumptions. Some PAC learning problems may also become computationally difficult in large and complex environments.

Q6. Explain reinforcement learning: learning from rewards, passive and active learning.

1. Reinforcement Learning is a branch of Artificial Intelligence where an agent learns by interacting with the environment. The agent performs actions and receives rewards or penalties based on its behavior.
2. The main objective of reinforcement learning is to maximize long-term rewards through continuous learning and experience. The agent gradually improves its strategy or policy over time.
3. Learning from rewards is an important concept in reinforcement learning where the agent learns from feedback received after actions. Positive rewards encourage good actions while negative rewards discourage wrong actions.
4. In reinforcement learning, the agent observes the current state of the environment before selecting an action. After performing the action, the environment changes and provides feedback in the form of rewards or penalties.
5. Passive Reinforcement Learning is a type of reinforcement learning where the agent follows a fixed policy. The agent only learns how good the states are and does not decide actions independently.
6. In passive reinforcement learning, the actions are predefined and the system estimates utilities of different states. Methods such as Direct Utility Estimation, Adaptive Dynamic Programming, and Temporal Difference Learning are commonly used.
7. Active Reinforcement Learning is a type where the agent learns which actions should be performed. The agent explores the environment and identifies the best policy on its own.
8. In active reinforcement learning, the agent performs exploration and exploitation to maximize rewards. Common algorithms used include Q-Learning and SARSA which help evaluate action quality.
9. Reinforcement learning is widely used in robotics, self-driving cars, game playing, recommendation systems, and automated control systems. These applications require intelligent decision-making in changing environments.
10. The major advantage of reinforcement learning is that systems can learn automatically without labeled training data. However, reinforcement learning may require large training time, high computational cost, and careful reward design.